How AI affects the U.S. Job Market?

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Summary of Questions and Results

- **Research Question #1**: From 2021 to 2024, do U.S. occupations with higher AI Occupational Exposure (AIOE) experience statistically different changes in employment levels than occupations with lower AIOE, after grouping occupations into high- vs low-exposure tiers (top vs bottom quartile of AIOE scores)?
  -> Yes. High-AIOE occupations experienced significantly faster employment growth than low-AIOE occupations between 2022 and 2024.

- **Research Question #2**: From 2021 to 2024, do U.S. occupations with higher AI Occupational Exposure (AIOE) experience statistically different changes in wages than occupations with lower AIOE, after grouping occupations into high- vs low-exposure tiers (top vs bottom quartile of AIOE scores)?
  -> Yes. While wage dispersion shrank across all occupations during this period, high-AIOE occupations experienced a significantly smaller decline than low-AIOE occupations (−3.03% vs. −4.01%).

- **Research Question #3**: Which occupations show the strongest "growth signals" from 2021–2024 in both employment growth and wage growth, and how are these high-growth occupations distributed across AIOE exposure levels (low/medium/high)?
  -> High-AIOE occupations are disproportionately represented among top performers. These top-performing, high-AIOE occupations are predominantly white-collar professional roles (e.g., Sales Managers, Civil Engineers, Medical and Health Services Managers) rather than explicitly AI or technology jobs.

Motivation

Artificial intelligence is rapidly changing how work is done in the U.S., not only by creating new AI-related roles but also by reducing roles that could be partially replaced by AI. For students and early-career workers, this creates a practical uncertainty: which occupations and industries are expanding, which are contracting, and how do those shifts relate to AI exposure?

This project uses authoritative labor market outcomes (like employment and wages) together with a published AI exposure measure to test whether more AI-exposed occupations/industries show different employment and wage changes than less-exposed ones during 2021–2024. The goal is to provide a transparent, data-driven view of which segments of the job market appear AI-complementary versus AI-risk. These findings can help new graduates prioritize career pathways.

Dataset

We will use two real, publicly available datasets:
Occupational Employment and Wage Statistics (OEWS) from the U.S. Bureau of Labor Statistics. OEWS provides annual U.S. estimates of employment levels and wage measures for detailed occupations and industries. We will use OEWS data for 2021–2024 to compute employment and wage changes by occupation and by industry.
Source: *Occupational Employment and Wage Statistics (OEWS) Tables*

Artificial Intelligence Exposure dataset (AIOE) from Felten, Raj, and Seamans. This dataset provides quantitative exposure scores for AIOE (AI exposure by occupation). We will merge these exposure scores with OEWS outcomes using the shared SOC identifiers to compare labor market trends for high- vs low-exposure groups.
Source: *GitHub - AIOE-Data/AIOE*

Challenge Goals

Challenge Goal #1: Multiple Datasets. We will combine multiple OEWS releases (2021, 2022, 2023, 2024) with the Felten, Raj, and Seamans's AI exposure measures (as two different tables). We will first standardize and concatenate the OEWS year files into a consistent 2021–2024 panel, then perform join/merge operations to attach AIOE to occupation outcomes via SOC codes for RQ1 and RQ2. This enables richer analyses than any single dataset alone by linking labor-market outcomes to AI exposure.

Challenge Goal #2: Result Validity. We used statistical testing to prove that the patterns found in our analysis were not simply due to random variation. After merging the OEWS datasets with AI exposure, we will apply statistical methods to evaluate the relationships between AI exposure and wages or employment levels. By interpreting the results of these statistical tests alongside the visualizations and summary statistics found in the analysis, we can determine if our results are statistically significant/meaningful. The addition of these tests strengthens the credibility of our results beyond just a descriptive analysis.

Modification: We have changed our #2 challenge goal from **New Library (Altair)** to **Result Validity**, because we found seaborn and matplotlib are more straightforward to implement and Result Validity is necessary to make our conclusions more robust.

Methods

EDA (Exploratory Data Analysis)

- **Loaded AIOE and OEWS (2021–2024)** and reviewed schemas, sizes, and key identifiers.
- **Checked data quality**. E.g., missingness and unusable/empty columns) in each dataset.

- **Merged OEWS with AIOE** and quantified unmatched rows.
- **Diagnosed mismatches** (expected OEWS aggregates vs. unmatched detailed occupations) and decided on exclusions.
- **Cleaned the data** by dropping unmatched occupations and removing non-numeric placeholder rows after numeric conversion.
- **Validated final per-year datasets** for consistency across years.
- **Summarized variables of interest with descriptive stats** (for both quantitative and categorical variables).
- **Visualized key patterns**. Such as AIOE vs. wages scatter; employment summed by AIOE bins.

Further data processing

- **Create derived variables used in all research questions**. Merge four yearly DataFrames into a single wide-format panel with one row per occupation and year-suffixed columns. Compute employment-weighted average employment for use as weights throughout the analysis. Assign AIOE exposure tiers (Low/Medium/High) using employment-weighted quartile cutoffs. Compute year-over-year and cumulative growth rates (2022–2024) for employment and wage percentiles. Compute wage dispersion ratios per occupation per year and their 2022–2024 growth rates. Aggregate employment-weighted average wages and dispersion ratios by AIOE tier and year.

Research Question #1: AIOE vs. employment change (2022–2024)

- **Compute employment growth rate per occupation** as the percent change from 2022 to 2024.
- **Group occupations by AIOE tier** (Low = bottom 25%, High = top 25%; Medium for the rest), where tier cutoffs are based on employment-weighted percentiles.
- **Compare employment growth across tiers** using group summaries (weighted and unweighted means).
- **Test for a statistical difference in employment change** between High vs Low AIOE occupations using both an unweighted and an employment-weighted one-tailed Welch t-test.

Research Question #2: AIOE vs. wage dispersion change (2022–2024)

- **Compute wage change per occupation** (absolute value and percent change).
- **Group occupations by AIOE tier** as RQ1.
- **Compare dispersion growth across tiers** using group summaries (weighted and unweighted means).
- **Test for a statistical difference in dispersion growth** between High vs. Low AIOE occupations using both an unweighted and an employment-weighted one-tailed Welch t-test.

Research Question #3: Identify “growth-signal” occupations and relate to AIOE

- **Compute both change metrics** (employment growth rate and wage dispersion growth rate) for each occupation from 2022 to 2024.

- **Assign each occupation a performance grade (1–5)** based on the combination of its employment growth tier and dispersion growth tier.
- **Identify the high-growth set** and summarize which occupations fall into each performance grade..
- **Describe distribution across exposure levels** by computing, for each performance grade, what percentage of its total employment and occupation count falls within each AIOE tier (H/M/L).

EDA results

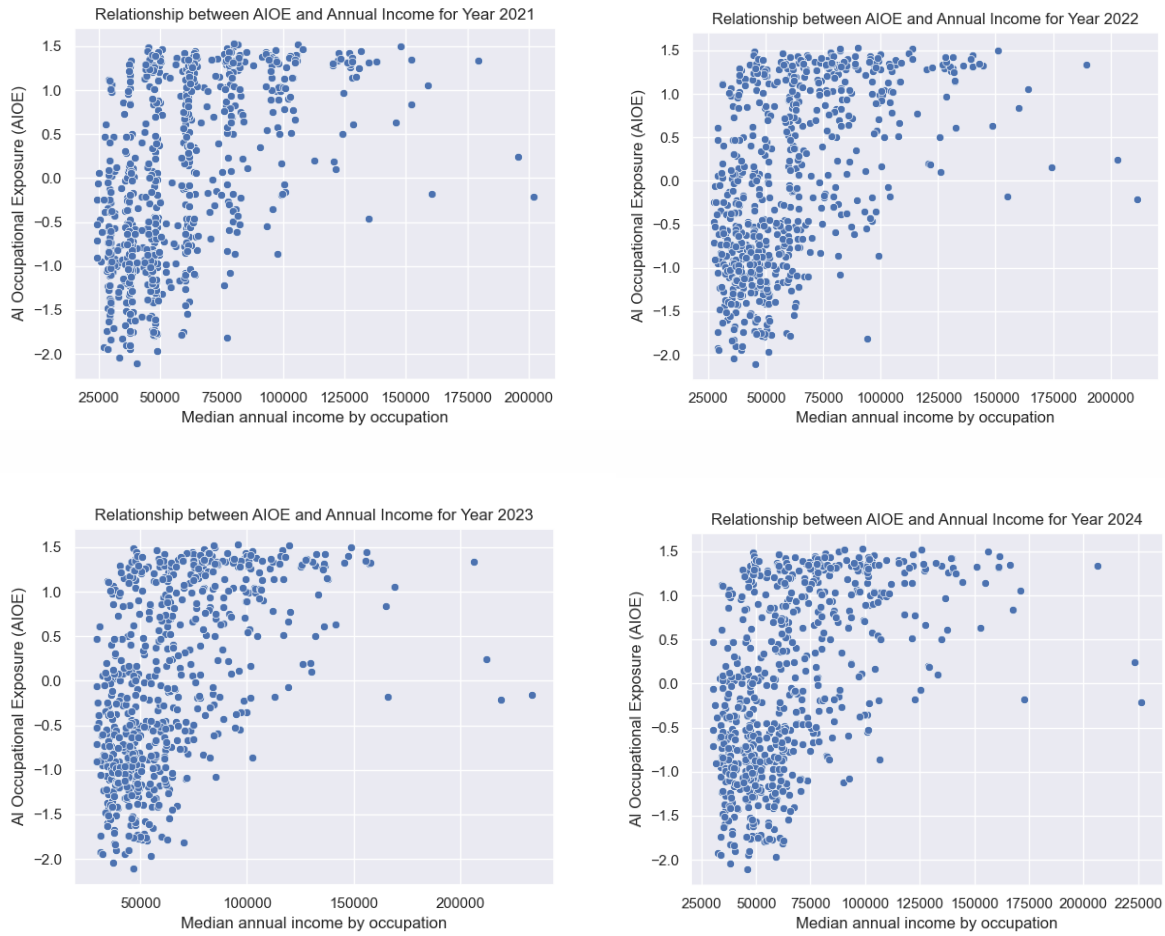
(A) Methods (please refer to “EDA” part in “Methods” section)

(B) Data structure and quality: The AIOE dataset contains 774 occupations with no missing values. The OEWS datasets each contain ~1,400 rows and 32 columns per year (2021–2024), with no missing values in the fields used in our analysis (aside from six nearly empty columns irrelevant to our research questions).

After merging, roughly 52% of OEWS rows lacked AIOE scores. This was largely expected: OEWS includes aggregated occupation groups (major, minor, broad) with no direct AIOE counterpart. Among detailed occupations, 158 out of 831 (about 19%, representing roughly 20% of total employment) remained unmatched. **Because this gap was consistent across all four years, it represents a systematic limitation we carry forward into all downstream analyses.** An additional 4–6 rows per year contained non-numeric placeholders in wage and employment fields and were removed after numeric conversion (less than 1%). The final cleaned datasets contained 668–669 occupations per year.

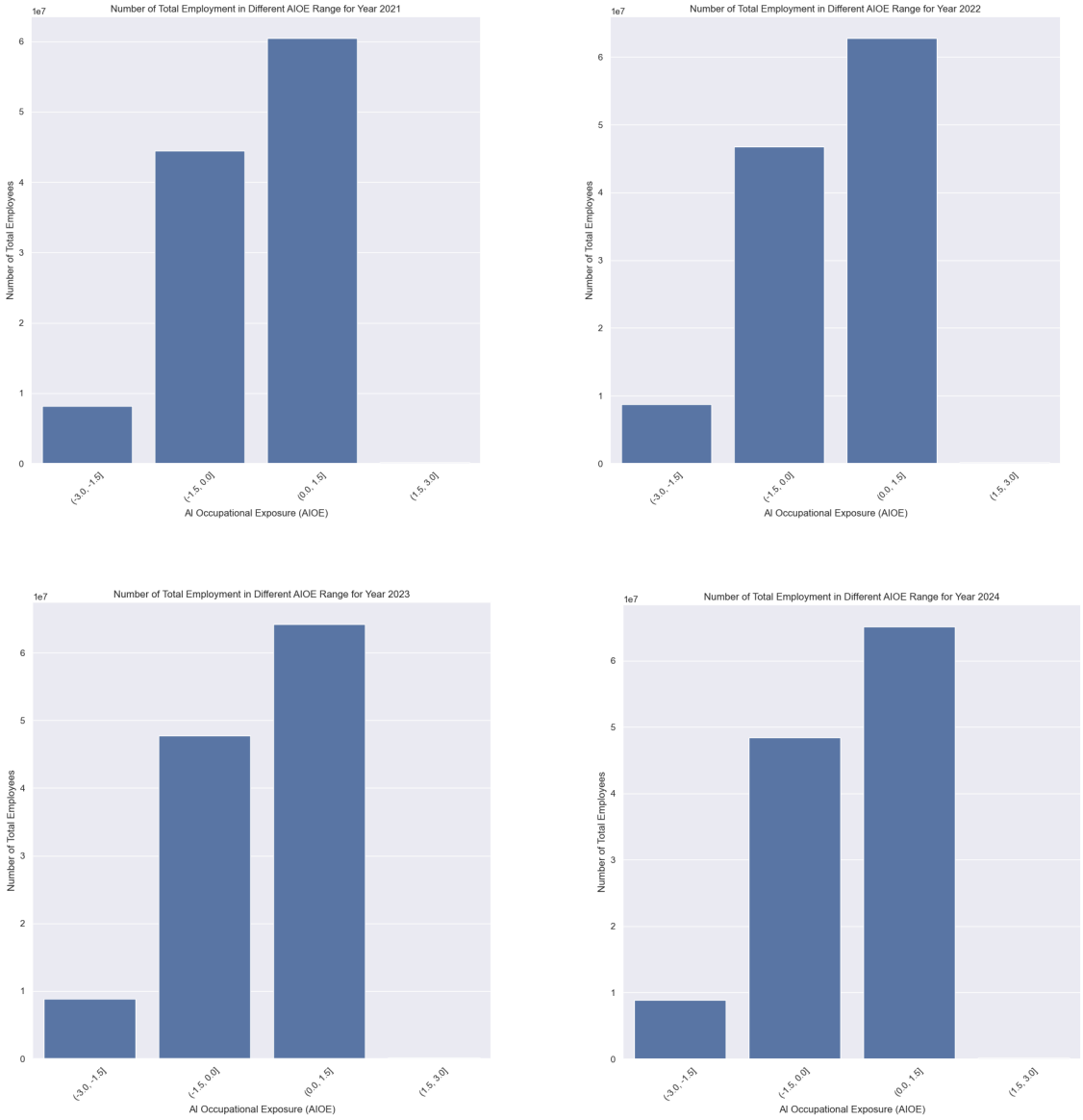
(C) Data descriptive patterns
AIOE is centered near zero with a standard deviation around 1, while employment and wages vary widely across 2021–2024. Mean employment per occupation ranges from roughly 170,000 to 184,000 across the four years, and median wages rise steadily from about $48,800 in 2021 to $59,100 in 2024. Each year's OCC_CODE values are unique, confirming that every row represents a distinct occupation.

(D) Visual insights: Two EDA visualizations directly informed our analytical design:
AIOE vs. median wage scatter (2021–2024): Higher-wage occupations tend to cluster at higher AIOE values, revealing a positive association between AI exposure and pay. This meant that comparing raw wage growth across AIOE tiers would be confounded by baseline salary differences, high-AIOE occupations already start higher. **This insight motivated our decision in RQ2 to use wage dispersion ratios rather than raw wage levels as the outcome variable**.



Caption: Higher-income occupations tend to show higher AIOE.

Total employment by AIOE bins (2021–2024): The bulk of U.S. employment is concentrated in middle AIOE ranges, with relatively few workers at the extremes. If we had used unweighted quartiles, the tier cutoffs would be driven by many small, niche occupations rather than reflecting where most workers actually sit. **This motivated our use of employment-weighted percentiles for all tier assignments (AIOE, employment growth, and dispersion growth), ensuring that the H/M/L groupings represent meaningful shares of the workforce**.

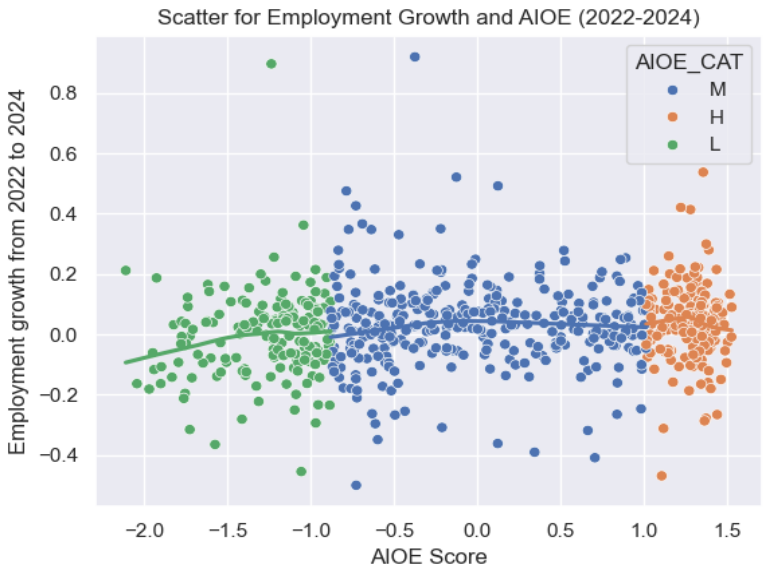


Caption: Employment is concentrated in moderately exposed occupations, with relatively little employment in the extreme high-exposure range.

Results

Note on data's time window: **All three research questions use 2022-2024 as the primary analysis window** rather than the full 2021-2024 span. Because after deep-diving into the data analysis for the research questions, we found that the 2021-2022 employment and wage changes were unusually volatile and probably attributed to post COVID-19 pandemic recovery dynamics. To get a cleaner signal, we **anchor all growth metrics to 2022 as the baseline year**.

(A) RQ1: High-AIOE occupations do outperform low-AIOE ones in terms of employment counts from 2022 to 2024.



Caption: High-AIOE occupations tend to have a higher unweighted average employment growth rate from 2022 to 2024.

Segment	Low-AIOE	Medium-AIOE	High-AIOE
Weighted average employment growth rate from 2022 to 2024	2.92%	3.59%	4.33%

We found that weighted and unweighted average employment growth rate from 2022 to 2024 favors high-AIOE occupations. So we first developed **a t-test on weighted average employment growth** to verify this trend from a macro workforce opportunity perspective.

- **Null hypothesis**: The weighted average of high-AIOE occupations' employment growth rates is equal to or less than that of low-AIOE occupations at 10% significance level.
- **Alternative hypothesis**: The weighted average of high-AIOE occupations' employment growth rates is greater than that of low-AIOE occupations at 10% significance level.

Outcome: the p-value for this test is 6.03%, which is less than the 10% threshold level. We reject the null hypothesis.

In addition, we developed **an additional t-test on unweighted average employment growth** to verify this trend from an occupation feature perspective to guarantee that we get a robust conclusion for new-graduates to decide safely which careers to pursue.

- **Null hypothesis**: The unweighted average of high-AIOE occupations' employment growth rates is equal to or less than that of low-AIOE occupations at 10% significance level.
- **Alternative hypothesis**: The unweighted average of High-AIOE occupations' employment growth rates is greater than that of low-AIOE occupations at 10% significance level.

Outcome: the p-value of this test is 0.13%, which is less than the 10% threshold. So we reject the null hypothesis.

Conclusion: Both tests point in the same direction:high-AIOE occupations grew faster than low-AIOE ones between 2022 and 2024. The unweighted test (p-value = 0.13%) produces a strong signal since it treats each occupation equally, giving more emphasis onto many relatively small-sized high-AIOE occupations that observed rapid growth. The weighted test (p-value = 6.03%) is more conservative, since it reflects how the overall workforce shifted.

The outcomes suggest that AI exposure is associated with stronger employment growth at both the individual occupation level and the aggregated workforce level. The effect is particularly notable given that 2022-2024 is the period that the mainstream emergence of large language models and generative AI tools.

(B) RQ2: In the era of shrinking wage dispersion, high-AIOE occupations tend to shrink slower than low-AIOE occupations during 2022-2024.

Segment	Low-AIOE	Medium-AIOE	High-AIOE
Aggregated percentile 25 (PCT25) wage growth rate from 2022 to 2024	11.45%	11.52%	10.79%
Aggregated median wage growth rate from 2022 to 2024	10.97%	10.22%	10.50%
Aggregated percentile 75 (PCT75) wage growth rate from 2022 to 2024	9.86%	8.97%	9.23%

Though the aggregated percentile 25 (PCT25) / median / percentile 75 (PCT75) wages of high-AIOE occupations grew slower than those of low-AIOE occupations from 2022 to 2024, that situation was probably caused by salary catching-up effect post COVID-19 pandemic. We can find that the overall average of PCT25 wage grew faster than that of median wage, which grew faster than that of PCT75, regardless of their AIOE categories.

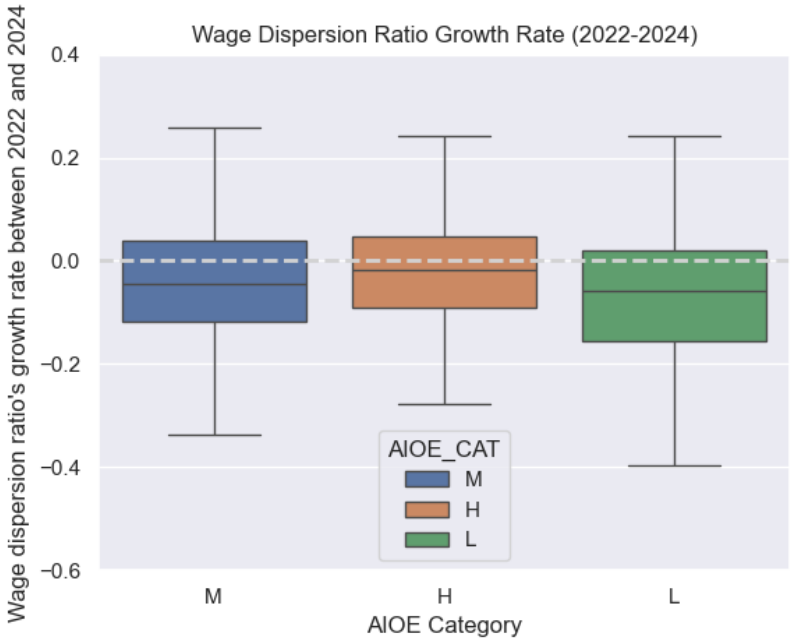
Segment	PCT25 wage	Median wage	PCT75 wage
Growth rate of aggregated wages from 2022 to 2024	11.38%	10.57%	9.35%

And high-AIOE occupations tend to be well-paid jobs, so we need to examine the effect from another lens.

We use wage dispersion ratio, which is (PCT75 wage - PCT25 wage) / median wage, to examine the effect that high-AIOE could help to increase or resist the shrinking of wage dispersion. We assume that people should be able to afford their families with the wages they get from their full-time employment (at least with a PCT25 wage of their occupations). The more wage dispersion ratio an occupation gets, the more affluent wage most people in this trade could earn. Therefore, a high wage dispersion ratio is a good indicator of good-paying jobs.

In addition, AI could assist people to increase their working efficiency, but it hardly can break through the upper limit of people's expertise (since they are already professionals in their trade). Therefore, with AI assistance, high-performing professionals tend to perform even better and more efficiently, and low-performing professionals may get replaced by the combinations of high-performing professionals and AI agents.

Wage dispersion ratio	2022	2024	Change	Growth rate
High-AIOE	56.98%	55.25%	-1.73%	-3.03%
Low-AIOE	39.35%	37.76%	-1.59%	-4.01%



Caption: High-AIOE occupations as a total tend to have higher 25-percentile, median, and 75-percentile wage dispersion growth rate from 2022 to 2024, compared with medium- and low-AIOE occupations.

We find that high-AIOE occupations' aggregated wage dispersion ratio shrank slower (or grew faster in the numerical sense) than low-AIOE ones from 2022 to 2024. So we developed **a t-test on weighted average dispersion growth rate** to verify this trend from a macro workforce perspective.

- **Null hypothesis**: High-AIOE occupations' weighted average dispersion growth rate is equal or less than that of low-AIOE at the 5% significance level.
- **Alternative hypothesis**: High-AIOE occupations' weighted dispersion growth rate is larger than that of low-AIOE at the 5% significance level.

Outcome: the p-value of this test is 4.8%, which is less than the 5% significance level. So we reject the null hypothesis.

Furthermore, we developed **an additional t-test on the unweighted average wage dispersion growth rate** to verify this finding from an occupation feature perspective to guarantee that we get a robust conclusion for new-graduates to decide which careers to pursue.

- **Null hypothesis**: high-AIOE occupations' unweighted average wage dispersion growth rate is less than or equal to that of low-AIOE at the 5% significance level.
- **Alternative hypothesis**: high-AIOE occupations' unweighted average wage dispersion growth rate is larger than that of low-AIOE at the 5% significance level.

Outcome: the p-value of this test is 1.6%, which is less than the 5% significance level. So we reject the null hypothesis.

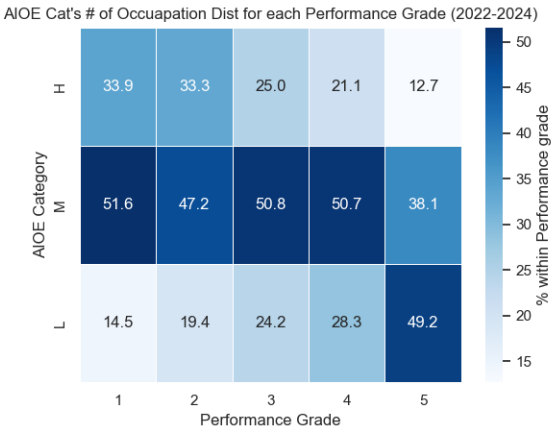
Conclusion: Both the weighted and unweighted tests confirm the same direction: high-AIOE occupations experienced a significantly smaller decline in wage dispersion than low-AIOE occupations did from 2022 to 2024. The difference in their shrinking speed (3.03% vs. 4.01%) is modest in absolute terms but statistically significant. The finding supports the interpretation that AI tools are amplifying the productivity and then compensation of top performers in high-AIOE occupations, moderating the compression of wage spreads that is occurring more broadly across the economy.

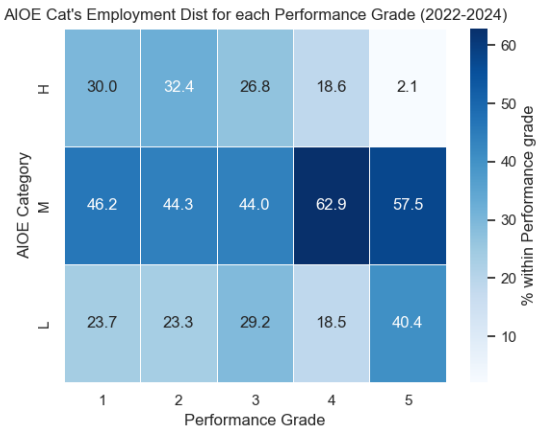
(C) RQ3: In terms of employment growth and salary dispersion growth rate, high-AIOE occupations account for more employment and occupation counts than low-AIOE ones do in good-performing groups during 2022 and 2024.

In addition to AIOE split, we split all occupations into high-, medium-, and low-employment growth and high-, medium-, and low-wage dispersion increase (both based on 2022-2024 data) and score the performance of occupations per following rules:

Category	Combinations of employment growth category and Wage dispersion growth rate category
1: Very good	HH (High employment growth and High wage dispersion growth rate)
2: Good	HM, MH
3: Average	MM, HL, LH
4: Poor	LM, ML
5: Very poor	LL

We found that High-AIOE occupations account for more employment and occupation counts than low-AIOE ones do in good-performing groups between 2022 and 2024.





Caption: In high grades (1 and 2), high-AIOE occupations outnumber low-AIOE ones in terms of both occupation count and employment, and undernumber or be similar to them in low grades (4 and 5).

Since medium AIOE occupations account for 50% of total employment while both high- and low-AIOE occupations account for 25% each, high-AIOE occupations actually perform better than both low- and medium-AIOE occupations in the “very good” and “good” groups.

Conclusion: This provides a clear directional signal: high-AIOE occupations are substantially more likely to be top performers and substantially less likely to be laggards on the combined employment growth and wage dispersion change metric.

Category	High-AIOE occupations
Very good	Sales Managers; Medical and Health Services Managers; Training and Development Specialists; Civil Engineers; Industrial Engineers; Construction Managers; Property, Real Estate, and Community Association Managers; Human Resources Managers; Aerospace Engineers; Social Science Research Assistants; Physical Scientists, All Other; Physicists; Foreign Language and Literature Teachers, Postsecondary; Compensation and Benefits Managers; History Teachers, Postsecondary; Psychologists, All Other; Administrative Law Judges, Adjudicators, and Hearing Officers; Nuclear Engineers; Social Work Teachers, Postsecondary; Emergency Management Directors; Environmental Science Teachers, Postsecondary

Notably, the high-growth, high-dispersion-growth, high-AIOE occupations are predominantly white-collar professional roles rather than explicitly AI or technology industry jobs. This is an important practical finding: benefiting from AI exposure does not require entering the AI industry directly. Instead, it suggests that professionals who possess deep domain expertise in fields and develop AI proficiency to be a productivity amplifier are well-positioned to see both employment and compensation growth.

The practical implication for new-graduates and early-career workers is to prioritize building domain expertise in their trades while applying AI tools to complement rather than replace their contributions.

Impact and Limitations

Impact: The project provides a data-driven perspective on how AI exposure relates to employment growth and wage dynamics in the U.S. labor market. By linking the AI Occupational Exposure (AIOE) dataset with Occupational Employment and Wage Statistics (OEWS), we identify patterns suggesting that occupations with higher AI exposure tend to experience stronger employment growth and slower declines in wage dispersion between 2022 and 2024.

These findings suggest that AI often acts as a productivity complement rather than a direct replacement in many professional occupations. For students and early-career workers, the results imply that building strong domain expertise while learning to leverage AI tools may improve long-term career prospects.

Limitations: Despite these insights, several limitations remain.
First, although we anchored growth metrics to 2022 to reduce volatility, some observed trends may still reflect post-COVID-19 economic recovery rather than purely AI-driven effects. Structural changes in the U.S. economy during this period; such as shifts in demand across industries and rapid adoption of new technologies may also influence employment and wage outcomes.

Second, our analysis relies on observational correlations rather than causal identification. While AI exposure is associated with certain labor market outcomes, the analysis cannot definitively establish that AI exposure directly causes these changes.

Work plan Evaluation

Our initial work plan was directionally accurate, but several steps required more time than expected. In particular, merging the AIOE and OEWS datasets required extensive troubleshooting because many OEWS rows represent aggregated occupation groups that do not have direct AIOE matches. As a result, we spent additional time diagnosing mismatches and defining consistent filtering rules across the 2021–2024 datasets.

Data cleaning also required more effort than anticipated. OEWS contains placeholder values in numeric fields that appear valid initially but become missing after numeric conversion, requiring additional filtering steps.

Overall, while the timeline underestimated the effort required for data preparation, the general structure of our plan; data collection, cleaning, analysis, visualization, and interpretation remained effective. Adjusting the schedule allowed us to complete the analyses and produce clearer visualizations and statistical tests aligned with our research questions.

Testing

To ensure the correctness and reliability of both the EDA and Analysis pipelines, we implemented unit and integration testing strategies. Unit testing used small synthetic DataFrames to check core functions, verifying aspects such as missing value handling, merge correctness, filtering after merging, column narrowing, and numeric coercion. As a result,

- All unit tests passed successfully.
- Integration testing was conducted on full datasets to validate loading and merging operations and to confirm that dataset sizes are correct, and that visualization files regenerate without errors.

This dual-layered approach ensures both logical correctness at the micro level and process reliability at the macro level, increasing our confidence that findings accurately represent the datasets.

Citations and Collaboration

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